

Junmo Lee

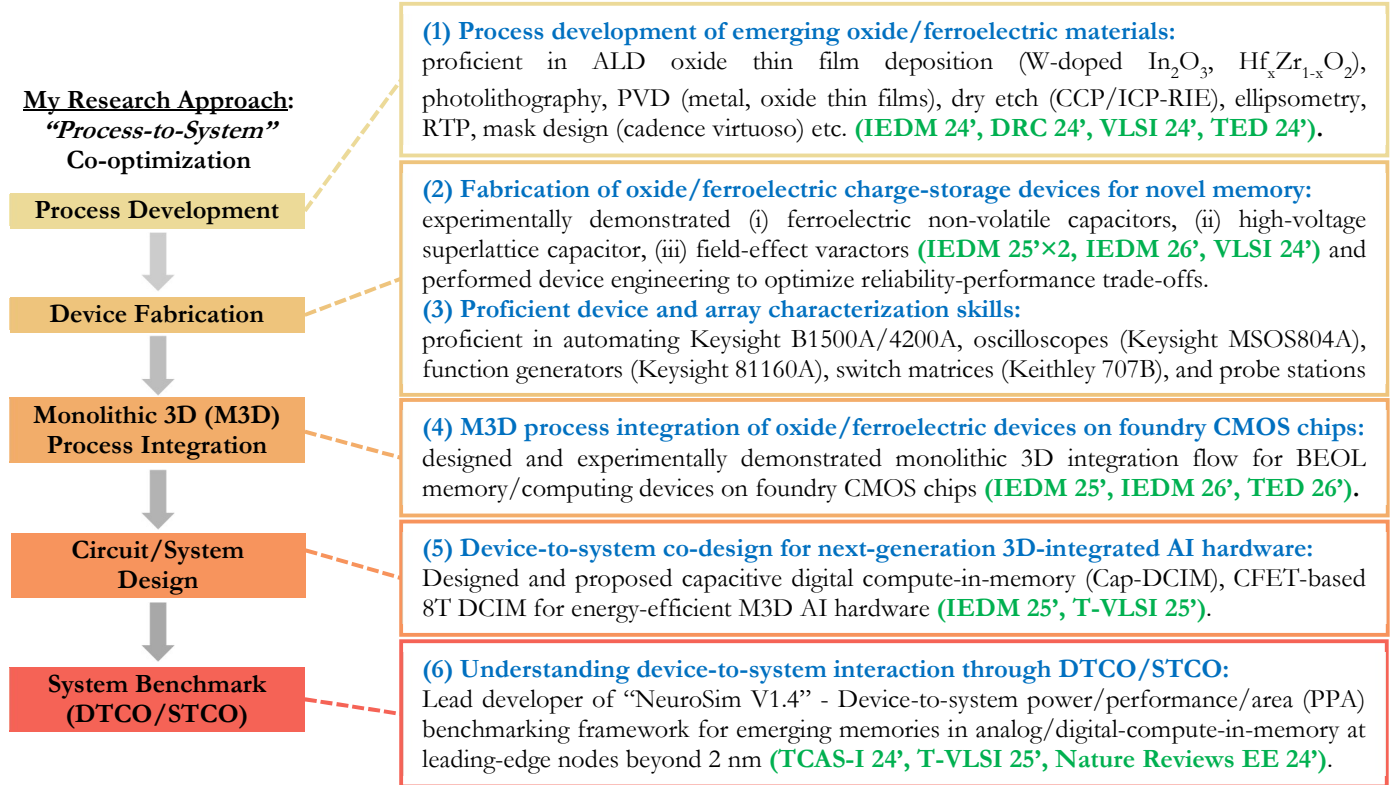
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SUMMARY

I am an **end-to-end device engineer** bridging process development and system-level innovations to translate emerging device concepts into actionable process flows for next-generation memory technologies. I have led cross-functional device research spanning **(i) Hands-on Process-Integration**: monolithic 3D process integration of TFTs and novel capacitive memories based on hafnia-based ferroelectric ($\text{Hf}_x\text{Zr}_{1-x}\text{O}_2$) and oxide semiconductor (W-doped In_2O_3) materials; **(ii) Memory Device Engineering**: optimization of reliability-performance trade-offs (e.g., retention vs memory window, TDDB vs capacitance density) in novel capacitive memory devices; **(iii) Device-to-System Codesign**: design of emerging-device-based next-generation memory/computing hardware (e.g., capacitive 3D DRAM, compute-in-memory, novel AI accelerators); and **(iv) Device-to-System Benchmarking**: understanding device-to-system interactions through end-to-end DTCO/STCO framework development (NeuroSim V1.4). The overview below summarizes my research experience and skills ((1)-(6)) from process to system, supported by my **first-author publications**.



→ With comprehensive device research skills spanning hands-on process integration to system-level evaluation, I aim to connect foundry technology development with system design to advance next-generation memory and computing.

EDUCATION

Georgia Institute of Technology (Georgia Tech)

Atlanta, Georgia

Doctor of Philosophy

August 2022 – August 2027 (expected)

Major in Electrical and Computer Engineering, Minor in Physics (currently 5th year Ph.D. Student)

SRC research scholar, Internship in Intel Logic Technology Development (LTD), focusing on 18A technology

Advisor: Dr. Shimeng Yu

Cumulative GPA: 3.91/4.0

Seoul National University (SNU)

Seoul, South Korea

Bachelor of Science

March 2016 - August 2022

Major in Electrical and Computer Engineering

Cumulative GPA: 3.95/4.3 (overall) 3.95/4.3 (major), graduated summa cum laude

HONORS & AWARDS

- (1) **Award Nomination (2025):** Roger A. Haken Best Student Paper Award at IEDM 25'
- (2) **Award Nomination (2024):** Best Oral Presentation Award at DRC 24' (top 15 finalist paper)
- (3) **Highlight paper (2025):** IEDM 25' work on "Monolithic 3D Integration of Dual-Gated ALD Oxide-Channel Non-Volatile Capacitive Memory on 40nm Si CMOS for Digital Compute-in-Memory" selected as conference highlight
- (4) **Media Coverage (2025):** IEDM 25' work featured in Nature Electronics (<https://www.nature.com/articles/s41928-025-01533-9>) and Semiconductor Engineering (<https://semiengineering.com/oxides-bring-low-leakage-transistors-to-leading-edge-memories/>)
- (5) **Gold Prize in IT Creativity Challenge (2017):** ranked 2nd in undergraduate student circuit design contest held by IEIE (The Institute of Electronics and Information Engineers)
- (6) **Excellent Tutor Award (2021):** awarded to excellent peer tutors of the semester at Seoul National University

SKILLS

- (1) **Languages:** English (professional working proficiency), Korean (native)
- (2) **Microfabrication:** ALD (Zr-doped Hafnia, W-doped Indium Oxide), Photolithography, PVD (metal, oxide thin films), Dry etch (CCP/ICP-RIE), Mask design tools (Cadence Virtuoso, KLayout), Ellipsometry, RTP (rapid thermal processing/annealing)
- (3) **Device simulation:** TCAD (Sentaurus, Silvaco)
- (4) **Device characterization:** Proficient with performing & scripting electrical characterization using Keysight B1500A, Keithley 4200A, oscilloscopes, function generators, switch matrices, and semi-automatic probe stations (Summit 12K). Includes device reliability (time-dependent dielectric breakdown) and device-to-device variability characterization to support failure-mechanism analysis.
- (5) **Circuit simulation:** Cadence Virtuoso ADE Explorer
- (6) **Programming/Software:** Python, C++, MATLAB, JMP
- (7) **Code development:** Lead developer of NeuroSim V1.4 - C++/Python-based chip-level benchmarking framework for emerging memory technologies at leading nodes (beyond 3 nm), now widely adopted by the device/circuit research community.

WORK EXPERIENCE

Intel Corporation, Logic Technology Development (LTD) Hillsboro, Oregon
Device Engineer/Data Scientist Intern August 2026 – November 2026

- Physics-based compact modeling of BEOL & Si backside metal layer parasitics and FEOL advanced logic transistors for 18A technology

SMDL (Advisor: Prof. Jong-Ho Lee @ Seoul National University, ECE) Seoul, South Korea
Research Intern January 2021 – August 2022

- Optimized sensing-material thickness and post-deposition annealing conditions for resistive and TFT-based gas sensors through device and process analysis as part of a senior design project.
- Developed a hardware-efficient weight-update scheme and a metric for evaluating update noise in synaptic-device-based compute-in-memory systems with abrupt conductance modulation.

NMDL (Advisor: Prof. Sangbum Kim @ Seoul National University, ECE) Seoul, South Korea
Research Intern January 2020 – January 2021

- Analyzed training dynamics in synaptic-device-based compute-in-memory systems with nonlinear conductance modulation and developed the Reverse Update Scheme to mitigate the resulting nonlinearity.

Army Missile Command Taean, South Korea
Republic of Korea Army May 2018 – January 2020

- Earned the Excellence Award (ranked first) in Encryption Contest
- Discharged with army soldier award.

PUBLICATION SUMMARY

**** Refer to PUBLICATIONS below for the referred papers ([1] [2].. etc.) associated with each project**

(1) Device fabrication and characterization of functional oxide thin-film capacitors and transistors

- Pioneered ferroelectric non-volatile capacitive memories with ALD oxide channel, and performed device engineering to optimize the trade-off between retention and capacitance memory window (IEDM 24' [4], IEDM 25' [1])
- Experimental demonstration of charge-domain in memory search using ferroelectric non-volatile capacitive array (IEDM 25' [3])
- First experimental demonstration and characterization of monolithic 3D BEOL integration of ferroelectric non-volatile capacitive memories on TSMC's 40 nm chip (IEDM 25' [1], TED 26' [5])
- Developed BEOL-compatible high-voltage superlattice capacitors and performed comprehensive reliability/electrical characterization, such as time dependent dielectric breakdown (TDDB) characterization to optimize reliability-capacitance density trade-offs (IEDM 23' [6], DRC 24' [12], TED 24' [10]).
- First experimental demonstration and characterization of BEOL-compatible DC-DC converters with oxide thin-film transistors and high-voltage superlattice capacitors (VLSI 24' [7])

- Fabricated and characterized a novel 3D DRAM device, the “Capacitive Gain Cell,” and developed its bit-cost-scalable process-integration scheme and array-level operation (IEDM 26’ [2])

(2) System-level modeling and benchmarking of emerging device technologies

- Developed NeuroSim V1.4: System-level power/performance/area (PPA) benchmarking framework for emerging device and compute-in-memory technologies at advanced technology nodes (TCAS-I 24’ [8], T-VLSI 25’ [9], Nature Reviews EE 24’ [11], DAC 24’ [16])

- Performed DTCO of Digital Compute-in-Memory at leading-edge technology nodes down to 0.5 nm technology node (TCAS-I 24’ [8], T-VLSI 25’ [9])

(3) Device-to-system codesign of AI hardware technologies

- Designed a novel AI accelerator (Capacitive Digital-Compute-in-Memory) based on experimentally demonstrated non-volatile capacitive memories on TSMC’s 40nm chip (IEDM 25’ [1])

- Designed a novel 8T Digital Compute-in-Memory based on 0.5 nm CFET technology (T-VLSI 25’ [9], APCCAS 24’ [14])

- Designed a novel weight update algorithm for AI accelerators based on analog synaptic devices (JXCDC 22’ [15])

PUBLICATIONS

* indicates equal contribution

[1] [Selected Publication] J. Lee, C. Zhang, S. Park, H. Park, T.-H. Kim, L. Liu, E. Ambrosi, M. Song, X. Bao, M.-F. Chang, S. Datta, S. Yu, “Monolithic 3D Integration of Dual-Gated ALD Oxide-Channel Non-Volatile Capacitive Memory on 40nm Si CMOS for Digital Compute-in-Memory,” **IEEE International Electron Devices Meeting (IEDM) 2025**, San Francisco, USA.

(highlight paper, nominated for the best student paper award)

[2] [Selected Publication] J. Lee, K. Lee, C. Zhang, T.-H. Kim, M. Tian, L. J.-C. Liu, M.-F. Chang, S. Datta, S. Yu, “Logic-Compatible Capacitive Gain Cell Monolithically Integrated on Si CMOS for Bit-Cost-Scalable 3D Digital Compute-in-Memory,” **IEEE International Electron Devices Meeting (IEDM) 2026**, accepted.

[3] [Selected Publication] O. Phadke*, J. Lee*, P.-K. Hsu, C. Zhang, A. Khan, S. Datta, S. Yu, “Array-Level Demonstration of Charge-Domain In Memory Search using Back-End-of-Line Compatible Ferroelectric Nonvolatile Capacitors,” **IEEE International Electron Devices Meeting (IEDM) 2025**, San Francisco, USA.

[4] [Selected Publication] J. Lee, C. Zhang, M. Shon, J. Read, S. Deng, O. Phadke, P. V. Ravindran, M. Tian, Y.-C. Luo, T.-H. Kim, A. I. Khan, S. Datta, S. Yu, “BEOL-compatible non-volatile capacitive synapse with ALD W-doped In2O3 semiconductor layer,” **IEEE International Electron Devices Meeting (IEDM) 2024**, San Francisco, USA.

[5] [Selected Publication] J. Lee, C. Zhang, S. Park, H. Park, T.-H. Kim, L. Liu, E. Ambrosi, M. Song, X. Bao, M.-F. Chang, S. Datta, S. Yu, “Dual-Gated ALD Oxide-Channel Non-Volatile Capacitive Memory Monolithically Integrated on Foundry 40 nm CMOS for Capacitive Digital Compute-in-Memory,” **IEEE Transactions on Electron Devices**, vol. 73, no. 5, pp. 3091-3101, 2026.

(invited paper as a top-ranked student paper at IEDM 2025)

[6] [Selected Publication] S. Deng, J. Kwak, J. Lee, K. A. Aabrar, T.-H. Kim, S. Kirtania, G. Choe, C. Zhang, W. Li, O. Phadke, S. Yu, S. Datta, “BEOL compatible oxide power transistors for on-chip voltage conversion in heterogenous 3D (H3D) integrated circuits,” **IEEE International Electron Devices Meeting (IEDM) 2023**, San Francisco, USA.

[7] [Selected Publication] S. Deng*, J. Kwak*, J. Lee*, D. Chakraborty, J. Shin, O. Phadke, S. G. Kirtania, C. Zhang, K. A. Aabrar, S. Yu, S. Datta, “Demonstration of on-chip switched-capacitor DC-DC converters using BEOL compatible oxide power transistors and superlattice MIM capacitors,” **IEEE Symposium on VLSI Technology and Circuits (VLSI) 2024**, Hawaii, USA.

(recognized as a top-ranked student paper at VLSI 2024)

[8] [Selected Publication] J. Lee, A. Lu, W. Li, S. Yu, “NeuroSim V1.4: Extending technology support for digital compute-in-memory towards 1nm node,” **IEEE Transactions on Circuits and Systems I: Regular Papers**, vol. 71, no. 4, pp. 1733-1744, 2024.

[9] [Selected Publication] J. Lee, M. Shon, F. Waqar, S. Yu, “3-D Digital Compute-in-Memory Benchmark With A5 CFET Technology: An Extension to Lookup-Table-Based Design,” **IEEE Transactions on Very Large Scale Integration Systems**, 2025, accepted.

[10] [Selected Publication] S. Deng*, J. Kwak*, J. Lee*, D. Chakraborty, J. Shin, O. Phadke, S. G. Kirtania, C. Zhang, K. A. Aabrar, S. Yu, S. Datta, “Monolithic switched-capacitor DC-DC voltage converters using BEOL-compatible oxide power transistors and superlattice MIM capacitors,” **IEEE Transactions on Electron Devices**, vol. 71, no. 12, pp. 7999-8006, 2024.

(invited paper as a top-ranked student paper at VLSI 2024)

[11] [Selected Publication] Anni Lu*, J. Lee*, T.-H. Kim*, M. Karim, R. Park, H. Simka, S. Yu, “High-speed emerging memories for AI hardware accelerators,” **Nature Reviews Electrical Engineering**, vol. 1, pp. 24-34, 2024.

[12] [Selected Publication] J. Lee, S. Deng, J. Kwak, M. Shon, S. Datta, S. Yu, “Optimization of backside of silicon-compatible high voltage superlattice capacitor for 12V-to-6V on-chip voltage conversion,” **IEEE Device Research Conference (DRC) 2024**, College Park, MD. (Nominated for the best oral presentation award)

- [13] S. Deng, H.-C. Chou, C. Zhang, J. Kwak, A. D. Asokan, E. Quezada, E. Sarkar, **J. Lee**, J. Shin, D. Chakraborty, V. Sathe, S. Yu, S. Datta, "High-efficiency BEOL-compatible on-chip switched capacitor DC-DC converter using planar and vertical oxide transistors with trench capacitors," **IEEE International Electron Devices Meeting (IEDM) 2025**, San Francisco, USA.
- [14] M. Shon*, **J. Lee***, S. Yu, "3D digital compute-in-memory benchmark with A5 CFET technology," **IEEE Asia Pacific Conference on Circuits and Systems (APCCAS) 2024**, Taipei, Taiwan.
- [15] **J. Lee**, J. Hwang, Y. Cho, M. K. Park, W. Y. Choi, S. Kim, J. H. Lee, "CRUS: A Hardware-Efficient Algorithm Mitigating Highly Nonlinear Weight Update in CIM Crossbar Arrays for Artificial Neural Networks," **IEEE Journal on Exploratory Solid-State Computational Devices and Circuits**, 8(2), 145-154, 2022.
- [16] A. Lu, **J. Lee**, Y.-C. Luo, H. Li, I. Young, S. Yu, "Digital CIM with noisy SRAM bit: A compact clustered annealer for large-scale combinatorial optimization," **ACM/IEEE Design Automation Conference (DAC) 2024**, San Francisco, CA.
- [17] S. Deng, J. Kwak, **J. Lee**, S. Yu, S. Datta, "BEOL-compatible on-chip DC-DC converters," **IEEE International Interconnect Technology Conference (IITC) 2024**, San Jose, CA.
- [18] G. Jeong, J. Kwak, M. Vadlamani, **J. Lee**, S. Yu, "Physics-Informed Neural Network-Assisted Compact Modeling for Process Variation in Non-Volatile Capacitive Crossbar Arrays," **IEEE International Reliability Physics Symposium (IRPS) 2026**, Tucson, AZ, USA, pp. 1-4.
- [19] D. Yu, H. Hwang, B. Oh, **J. Lee**, S. Yu, T.-H. Kim, H. Kim, "A Unified Flash Memory Platform for Mode-Adaptive and Robust AI Computation," **Advanced Science**, e76619, 2026.
- [20] J. Kwak, G. Choe, **J. Lee**, S. Yu, "Monolithic 3D transposable 3T embedded DRAM with back-end-of-line oxide channel transistor," **IEEE International Symposium on Circuits and Systems (ISCAS) 2024**, Singapore.
- [21] J. Kwak, S. Deng, **J. Lee**, S. Datta, S. Yu, "Backside active power delivery with hybrid DC-DC converter enabled by amorphous oxide semiconductor transistors," **IEEE Transactions on VLSI Systems**, 2025.
- [22] J. Kwak, S. Deng, C. Zhang, G. Jeong, O. Phadke, S.G. Kirtania, **J. Lee**, S. Datta, S. Yu, "Machine Learning-Assisted Compact Modeling of AC Stress-Induced Bias Temperature Instability in Top- and Bottom-Gate W-Doped In2O3 Channel Semiconductor Transistors," **IEEE Transactions on Electron Devices**, 2025.
- [23] O. Phadke, P.-K. Hsu, Y.-C. Luo, J. Sharda, V. Garg, J. Read, **J. Lee**, H. Mulaosmanovic, S. Dunkel, S. Beyer, S. Yu, "Ternary content addressable memory with ferroelectric non-volatile capacitor," **IEEE Electron Device Letters** 2025.
- [24] J. Kwak, C. Zhang, G. Jeong, O. Phadke, S. G. Kirtania, **J. Lee**, S. Datta, S. Yu, "Machine learning-assisted modeling of AC stress-induced bias temperature instability in oxide channel transistors for 2T gain cell eDRAM," **IEEE International Reliability Physics Symposium (IRPS) 2025**, Monterey, CA.
- [25] M. Vadlamani, J. Jia, T. Xie, Y.-C. Luo, **J. Lee**, S. Li, S. Yu, "Capacitive Crossbar Array for Solving Matrix Equations in One-Shot," **IEEE Electron Device Letters**, vol. 46, no. 3, pp. 389-392, 2025.
- [26] W. Li, **J. Lee**, S. Yu, "Optimization strategies for digital compute-in-memory from comparative analysis with systolic array," **IEEE International Conference on Artificial Intelligence Circuits & Systems (AICAS) 2023**, Hangzhou, China.
- [27] W. Li, X. Zhang, **J. Lee**, F. L. Degertekin, S. Li, S. Yu, "Enabling ultra-low power ultrasound imaging with compute-in-memory sparse reconstruction accelerator," **IEEE Biomedical Circuits and Systems Conference (Bio-CAS) 2023**, Toronto, Canada.
- [28] W. Shin, B. Choi, **J. Lee**, S. Hong, Y. Jeong, G. Jung, J. Park, D. Kim, C. Lee, J. H. Lee, "Sub-ppm Detection of NO2 gas using IGZO TFT-type Gas Sensor with Optimal Signal-to-Noise Ratio", **International Conference on Solid State Devices and Materials (SSDM) 2021**, virtual.

Under Review

- [29] **J. Lee**, C. Zhang, S. Datta, S. Yu, "Pulse-Mode Operation and Reliability of BEOL-Compatible Ferroelectric Non-Volatile Capacitive Memories with Amorphous Oxide Semiconductor Channels," **IEEE Transactions on Nanotechnology**
(journal extension paper of IEDM 2024)

TEACHING EXPERIENCE

Undergraduate Tutor (@ Seoul National University)

Peer Tutor (Semiconductor Devices)

Seoul, South Korea

September 2021 – December 2021

- Provided 30 hours of online lecture, received Excellent Tutor Award

Undergraduate Tutor (@ Seoul National University)

Peer Tutor (Introduction to Electronic Circuits and Laboratory)

Seoul, South Korea

September 2022 – December 2022

- Provided 24 hours of online lecture

SCHOLARSHIPS

(1) Merit-based Scholarship, Seoul National University (2020): provided to undergraduate students with excellent academic record

(2) Scholarship, Korea Electric Power Corporation (2021): provided to undergraduate students with excellent academic record